

- 2) What in your view would be a desirable future cropping pattern for India?

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- 3) Highlight the different issues related to water availability and its use in India.

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4.5 BIODIVERSITY

India contains a great wealth of biological diversity in its wetlands and in its marine areas. There are about 350 species of mammals, 1,224 species of birds, 408 species of reptiles, 197 species of amphibians, 2,546 species of fishes, and 15,000 flowering plants. The importance of these biological resources cannot be over emphasised for the continued wellbeing of Indian population.

A large number of both flora and fauna are faced with threat of extinction because of the increasing demand being made on natural resources.

The government has responded by taking several measures including legislative measures, among which the more important is the Biological Diversity Act, 2002.

4.5.1 The Biological Diversity Act, 2002

The Act covers conservation, use of biological resources and associated knowledge prevalent in India for commercial or research purposes or for the purpose of bio-survey and bio-utilisation. It provides a framework for access to biological resources and sharing the benefits arriving out of such access and act. The Act also includes in its ambit the transfer of research results and application for intellectual property rights relating to Indian biological resources.

4.6 FOREST RESOURCES

Forests produce the requisite raw materials for industries, defence, communications, domestic use, and other public purposes. They contribute to the country's exports and create a large volume of employment in the primary, secondary, and tertiary sectors. They also provide materials like fuelwood, small timber, fodder, etc. for direct use by the agriculturists. The benefits from forests in the matter of soil conservation, recreation, wildlife, etc. have been well-recognised.

4.6.1 Present Position

Forests occupy about 807.3 lakh hectares or about 24.5 per cent of the total geographical area. (Of this dense forest category, *i.e.*, 40 per cent or more of the area covered by trees, amounts to about 58.0 per cent.) About 433 lakh hectares or 61.0 per cent are exploitable; another 178 lakh hectares or about 25 per cent are potentially exploitable. Among the States, Madhya Pradesh has the maximum forest area of 77,265 sq.km., followed by Arunachal Pradesh (68,045 sq.km.) and Chhattisgarh (56,448 sq.km.).

The area under forests in India is low not only as compared to the forest area in countries like Japan (64 per cent), Sweden (66 per cent), South Korea (63 per cent), Canada (27 per cent), and USA (25 per cent), but is also much less than the norm of 33 per cent of the total reported area recommended in the National Forest Policy of 1952. The per capita forest land in India is 0.06 hectares as against the world average of 2.06 hectares. Further, the productivity of Indian forests is very low: 1.2 cubic meters per hectare per year as against the world average of 2.1 cubic meters.

Forests in India are still under huge pressure and are shrinking over time. Firstly, the rate of forest land diverted towards development projects has been happening at an unprecedented rate. But this diversion also happens because there is no value seen in forests—other than the cost that has to be paid for diversion of land by the project proponent. Instead, there is value in the dam, road or mine for which the land is needed. So, this pressure on forest land is bound to increase. We must also note that forests are the last remaining public lands in the country and the acquisition of private land will become even more expensive and contentious in future.

Secondly, the compensatory plantation done to replace the original, natural forests during diversion of forestlands for projects have so far yielded no impactful results. Some of these new areas were even earmarked for other projects or expansion of existing ones, even though they remain classified under government's Recorded Forest Area (in terms of legal status).

Thirdly, forests are under pressure to meet local needs and on account of illegal extraction. Today, it is an inconvenient truth that the poorest people in India live in the country's richest forests. The management of this green wealth has not brought any benefits to the locals. Amid all this, while deforestation and forest diversion will grow, we do not have any viable strategies for re-greening these lands. Thus, we will lose them gradually.

So how do we change this? *One*, we need to urgently value the economic, ecological and livelihood potential of forests and to incorporate this into national accounts. We need a robust methodology to bring the tangible (what we can measure) and intangible costs together. Though there is much talk about green accounting, the methodology is weak. For instance, there is no real assessment of minor (non-timber) forest produce. Other assessment of the contribution of forests to livestock or the hydropower sector is inflated or non-existent.

Two, we need to use this methodology to pay for standing forests. The 12th and 13th Finance Commissions allocated funds for standing forests, but this is a pittance. We, then, need states to transfer payment for standing forests—protected

for biodiversity, watershed, or other purposes– to local custodians. This will build local economies and local support for forest protection.

Three, at least 60 per cent of districts in India are affected by forest fires each year, and the top 20 districts in terms of fire frequency are located mainly in the Northeast, a joint report by the Ministry of Environment and Forests and Climate Change (MoEFCC) and World Bank has brought that out.

The top 20 districts in terms of area affected by fire from 2003 to 2016 account for 48 per cent of the total fire-affected area, the report found. In line with other parts of the world, people are the main driver of fires in India and forest fires are distributed close to people and infrastructure. Forest fires contribute to climate change by releasing carbon stored in trees, undergrowth, and soil into the atmosphere.

We need to use robust accounting methodology to increase the productivity of the remaining forest land. But we know that the business of cutting and planting trees that survive cannot be successful without the engagement of people who live in the forest. This has to be seen as the new opportunity for employment and economic growth. The way ahead is to build inclusive economies using green wealth.

4.6.2 National Forest Policy

The national forest policy was first enunciated in 1952 and subsequently revised in 1988. The important distinguishing features of this policy are as follows:

- 1) The policy lays emphasis on the conservation of forests and meeting the requirements of the tribal and rural people. The Scheduled Tribes and Other Forest Dwellers Act, 2006 has been enacted in pursuance of this policy.
- 2) Tribals will also be associated with the protection, regeneration and development of forests, and cooperatives run by them or by the Government will replace the present contractor system which has resulted in unchecked devastation of reserved forests.
- 3) Realising the fact that uncontrolled expansion of forest-based industries would adversely affect the conservation objective, the revised policy has made the following stipulations:
 - No forest-based industry, except in the small-scale and cottage sectors, would be permitted unless sustained availability of raw material is ensured.
 - As far as possible, a forest-based industry should produce its own raw material requirements.
 - Natural forests would not be made available to industries for undertaking plantations and for other objectives.
 - Forest produce would not be supplied to industries at concessional rates.

The initial results of the new policy have been mixed. Under the Green India Mission, it is proposed to increase forest cover by 5 million hectares and improve the quality of forests on another five million hectares over the next 10 years. The Green India Mission is one of eight missions under the National Action Plan on Climate Change. It aims to increase forest cover on 5 million hectares (ha) of